PYRIDOSTIGMINE

Myasthenia gravis
Cholinergic crisis

CHEMICAL NAME 3-[(dimethylcarbamoyl)oxy]-1-methylpyridinium bromide
CLASS Reversible acetylcholinesterase inhibitor
TRADE NAMES Mestinon · Regonol · Distinon

THERAPEUTIC USES / INDICATIONS

- Myasthenia gravis (symptomatic treatment).
- Reversal of non-depolarising neuromuscular block (IV, less common).
- Postural orthostatic tachycardia syndrome (off-label); paralytic ileus (rare).

ADMINISTRATION

- **PO** at regular intervals 30–60 min before meals (helps chewing/swallowing).
- MR tablets for overnight symptoms.
- Timing is important — doses may be needed every 3–4 h.

DOSAGES

- **Myasthenia:** 30–60 mg every 4–6 h; usual 300–1,200 mg/day in divided doses (max 1.8 g).
- **Children:** 1 mg/kg every 4 h initially.
- Lower doses in renal impairment.

PHARMACOKINETICS

- **Onset:** 30–60 min. **Duration:** 3–6 h (longer than neostigmine).
- Quaternary — poor oral absorption (10–20%), no CNS effects.
- Excreted in urine. **t½:** 1.5–2 h.

PHARMACODYNAMICS

- Inhibits **acetylcholinesterase** at neuromuscular junction → ↑ ACh to compete with autoantibodies at reduced receptors → ↑ muscle strength.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Abdominal cramps, diarrhoea, nausea.
- Increased salivation, sweating, lacrimation.
- Muscle fasciculations, cramps.

ADVERSE EFFECTS (SERIOUS)

- Cholinergic crisis (overdose): increasing weakness, respiratory failure, bradycardia, miosis, excessive secretions — hard to distinguish from myasthenic crisis.
- Bradycardia, AV block; bronchospasm.

CONTRAINDICATIONS & CAUTIONS

- Mechanical intestinal or urinary obstruction.
- **Caution:** asthma, bradycardia, recent MI, peptic ulcer, epilepsy, parkinsonism.

KEY INTERACTIONS

- **Aminoglycosides, fluoroquinolones, macrolides, magnesium, β-blockers:** worsen myasthenia.
- **Anticholinergics (atropine):** antagonise muscarinic effects (used for side effects).
- **Suxamethonium:** prolonged block.

NURSING MANAGEMENT

- Give on time; record muscle strength, swallowing, speech, ptosis before and after doses.
- Monitor respiratory function (vital capacity, single-breath count) during exacerbations.
- Recognise crisis: weakness + secretions + bradycardia (cholinergic) vs weakness alone (myasthenic).
- Check new drugs for myasthenia-worsening effects.

PATIENT EDUCATION

- Take doses on time; time doses 30–60 min before meals.
- Seek urgent help for breathing or swallowing difficulty.
- Carry a myasthenia alert card; check new medicines with pharmacist.
- Report diarrhoea, cramps, excess saliva.

KEY POINTS

- Anticholinesterase for myasthenia gravis.
- Cholinergic vs myasthenic crisis; both cause weakness.
- 30–60 mg every 4–6 h, before meals.
- Avoid aminoglycosides, magnesium.